

NETWORK INFRASTRUCTURE ASSESSMENT: A BASIS FOR ENHANCED NETWORK DESIGN

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ABSTRACT - This study presents an assessment and proposed upgrade of the network infrastructure at Cauayan City Hall with the goal of addressing existing hardware and software limitations and improving overall performance. The current network setup which includes insufficient printers, outdated routers, limited-port switches, and varying computer specifications, was found to contribute to inefficiencies and performance bottlenecks. The proposed design introduces a more robust system, incorporating advanced routers, managed switches, high-capacity printers, and strategically placed access points to enhance connectivity and support both wired and wireless devices. Additionally, the implementation of a tree topology network structure is recommended to replace the current star topology, improving scalability, reliability, and performance. Software updates, including enhanced security and productivity tools, will complement the hardware upgrades. Network performance will be optimized through the use of fiber-optic cabling, Wi-Fi 6 technology, VLANs, and DHCP for IP management, ensuring better bandwidth distribution and minimizing congestion. The study emphasizes the importance of network security through port security and VLAN configurations, and highlights the need for skilled personnel to manage and maintain the upgraded system. Overall, the proposed network infrastructure is designed to meet the growing demands of City Hall's operations, ensuring reliable and scalable connectivity, supporting future expansion, and enhancing overall network performance.

Keywords: Network Infrastructure, Network Design, Network security

1. INTRODUCTION

Every day a lot of transaction happens at Cauayan City Hall such as Business Permit, Licensing, Legal Services, Accounting, and more. The resilience of network infrastructure within public institutions has become a key foundation for guaranteeing efficient service and protection of sensitive data. As digital interfaces become more integrated into the day-to-day operation of public services, the importance of secure, dependable, and efficient network infrastructure cannot be understated.

be conducted in response to events such as security incidents or significant network changes. Currently, City Hall's primary Internet Service Provider (ISP) is PLDT, which is crucial in delivering the necessary connectivity and network services for its operation.

In today's digital age, the efficiency of city operations heavily relies on the robustness of their IT infrastructure to support their wide range of services. Over time, the demands on this infrastructure have increased due to the growing reliance on digital services and the

City Hall's network infrastructure assessment policy mandates unscheduled assessments to need for real-time data access and processing. However, without regular assessments and upgrades, network infrastructures can become outdated, leading to inefficiencies, security vulnerabilities, and disruptions in service delivery.

Among such institutions, City Hall stands out as a prominent administrative institution with a critical role in governance and service delivery to the community it serves. On 23-25 March 2023 City Isabela is awarded to be the first smart city. Being able to declare a smart city means having efficient and good networks and services with the utilization of digital solutions for the benefit of citizens and businesses (PCHRD, 2022). This study can contribute to maintaining the title of being a "Smart City" as risks and vulnerabilities in the network can be present day-to-day. Also, smart cities require high-speed, reliable communication networks to transmit data between devices, systems, and users.

Currently, City Hall faces challenges with its network performance, including slow data transfer rate, poor internet connection, and frequent downtime that causes delays in administrative processes and hinders the ability of the employees to serve the public efficiently. One major concern observed by the researchers is the physical security of network hardware, such as routers and servers. These critical components are not adequately secured, making them vulnerable to tampering, theft, or unauthorized access. Without proper physical security measures, the risk of physical breaches can lead to severe disruptions in network operations and compromise the integrity and confidentiality of sensitive data.

Addressing these problems is crucial to ensure the smooth and secure operations of City Hall. This assessment is essential in identifying vulnerabilities in the current setup including the lack of physical security for

network hardware, and provide a guide for improvements.

According to the study, physical assets within a critical infrastructure system is very crucial for enhanced security and systems dependent to it. This is particularly important for communication systems. This paper presents a multilayer complex network framework that takes into account the heterogeneity of the redundant infrastructure. (Herrera et. al, 2023) However, even this study contributes a lot in managing network infrastructure existing models do not sufficiently incorporate the concept of redundancy, which is crucial for network resilience and reliability. This study often utilizes a single-layer network approach, which fails to capture the complexity and heterogeneity of real-world communication infrastructures. To fill those limitations this study will implement complex network to model physical and logical layers of the network.

Here, this study intends to conduct an extensive assessment of the current network infrastructure of City Hall and propose an enhanced network design, improve network performance with a particular emphasis on identifying any loopholes that may expose the institution to cyber-attacks, operational inefficiencies, or both. The digital era, defined by the rapid rate of technological advancement and the evolving sophistication of cyber threats necessitates that public institutions such as City Hall not only stay current with these changes but also safeguard against potential vulnerabilities within their network system and aim to be a model of the first smart city here in Isabela.

2. REVIEW OF RELATED LITERATURE AND STUDIES

Design and Simulation of Local Area Network Using Cisco Packet Tracer

Simulation tools offer a way to predict the impact of a hardware upgrade, a change in topology. So, in this paper, a LAN network is designed using Cisco Packet Tracer. This study provides an insight into various concept such as topology design and simulation.

A range of studies have explored various aspects of local area network (LAN) design and simulation using Cisco Packet Tracer. G.K. (2020) focuses on performance evaluation and network segmentation, particularly examining the implementation of VLAN technology. The study by G.K. also delves into the configuration of both wired and wireless LANs, emphasizing the setup of DHCP, HTTP, and HTTPS protocols, alongside Ethernet and Wi-Fi technologies.

Building upon these concepts. Hossain (2019) finally provides a practical application by detailing the simulation and design of a university area network scenario using Cisco Packet Tracer, contributes to this body of work by giving an insight on how to properly use Cisco Packet Tracer and how crucial simulation process in building a well-designed network infrastructure (Thu, 2020)

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Assessment of Network Infrastructure

According to the study, the maintenance of system flow is critical for effective network operation. Any disruption to network facilities, whether affecting arc or nodes, poses a risk of service loss, potentially leaving users without

access to crucial resources. This study has proven that assessment of network infrastructure is an important way in identifying vulnerabilities in disrupting efficient flow in network (Matisziw et.al 2019) According to the study, critical infrastructure (Cis) provides essential services to the society it highlights the need for better management of networks interactions and independencies. Interdependencies cause cascading failures and, creates negative impact due to these failures. This study developed an integrated resilience assessment framework to identify and map interdependency-induced vulnerabilities in critical infrastructure networks (Hajializadeh, 2020)

According to the study, unforeseen disaster risk infrastructure system is expected to be resilient. Because of those inefficiencies, a resilience- based recovery assessment framework is proposed. The framework aims to find the most effective recovery when network is subjected to attacks (Yodo et.al,2020)

Community Networks:

According to the study, community detection is a fundamental problem in social network analysis consisting, roughly speaking, in unsupervised dividing social actors (modeled as nodes in a social graph) with certain social connections (modeled as edges in the social graph) into densely knitted and highly related groups with each group well separated from the others. Classical approaches for community detection usually deal only with the structure of the network and ignore features of the nodes (traditionally called node attributes), although the majority of real-world social networks provide additional actors' information such as age, gender, interests, etc. It is believed that the attributes may clarify and enrich the knowledge about the actors and give sense to the detected communities (Chunaev, 2020).

The interconnection of network architecture and channels of information transmission and storage in the process of the internet communication. The concept of digital folklore as compared to traditional folk culture is explained. Genre classification of digital folklore is introduced. The mechanisms of digital folklore functioning within different patterns of modern network architecture are analyzed. The idea of the research is introduced based on the comparative study. The article argues that the internet forms a special communicative environment that gives rise to innovative cultural and linguistic phenomena. It reflects users' thoughts and ideas, expressed in the form of a text, synthesized with graphics, sound, and animation (Melnikova et al., 2020)

LAN Infrastructure Design:

The network infrastructure design refers to the strategic development of both physical and logical elements within a computer network. This encompasses hardware, software, protocols, and configurations essential for facilitating communication and data transfer among devices and systems. Creating an effective network infrastructure design is paramount to meeting specific standard such as performance, scalability, security, and reliability, designed to the intended use case.

A study introduces an innovative network infrastructure design for data center networks, employing a hierarchical architecture. Here, the core layer handles data packet forwarding, the aggregation layer consolidates traffic, and the access layer facilitates connectivity to servers and storage devices. Additionally, the design integrates software-defined networking (SDN) and network function virtualization (NFV) to enhance network efficiency and diminish power consumption. The research illustrates that this design markedly diminishes network

latency and enhances network throughput in contrast to conventional data center network designs (Yang et al., 2020).

Network infrastructure devices, which facilitate data movement and communication, are vital components of modern networks. In wireless networks, the distance between stations and access points can impact Quality of Service (QoS) measurements, particularly concerning data distribution delays. Ghozali et al. (2020) demonstrated that in scenarios where wireless sensors transmit data (such as pH data acquisition) from node stations to web services, delays in data transmission can significantly affect measurement accuracy and the presentation of data on cloud servers.

to mobile services in underserved areas. In this context, the use of aerial radio access networks (ARANs) is a promising strategy to complement existing terrestrial communication systems. Involving airborne components such as unmanned aerial vehicles, drones, and satellites, ARANs can quickly establish a flexible access infrastructure on demand. ARANs are expected to support the development of seamless mobile communication systems toward a comprehensive sixth generation (6G) global access infrastructure. Dao et al., (2021)

LAN Security

Securing a local area network (LAN) infrastructure has become crucial due to the increasing issue in cyber-attacks.

This study proposes a Software Defines Networking (SDN) framework for LANs. The aim of the study is to optimize network management, improve performance, enhance security, and achieve cost effectiveness in LAN environments (Udo, Edward et.al,2020)

This study proposes a TorVPN access point as a solution for bolstering security and privacy on LANs. This solution leverages the Tor network to encrypt user traffic and incorporates a virtual private network (VPN) to offer supplementary security measures.

Nonetheless, the solution faced connectivity challenges attributed to the inherent characteristics of the Tor network (Osman, Mohd et al. 2021).

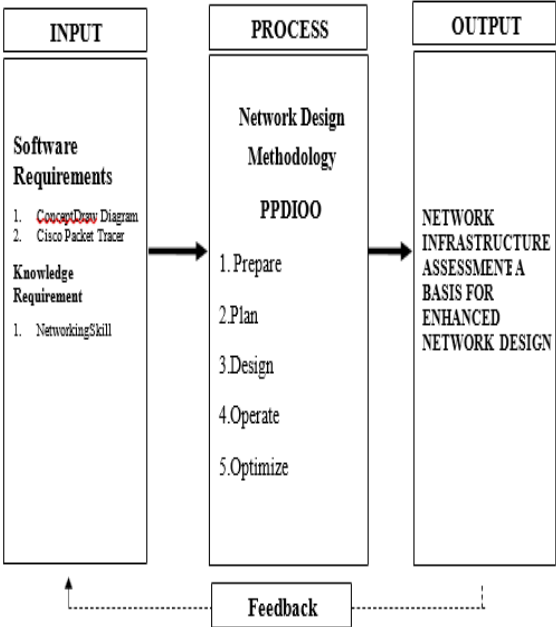


Figure 1. Research Paradigm of Network Infrastructure Design in Cauayan City Hall

This diagram presents the processes that illustrate the input, process, and output involved in identifying the objectives and capabilities of Network Infrastructure Design in City Hall. These processes include the input, which represents the requirements and approaches needed to fulfill the objectives, such as software requirements which include the Cisco Packet Tracer. Last but not the least, the knowledge in networking by the researchers followed by the process stage follows the PPDIOO Methodology that involves preparation, planning, designing, operating, and optimizing. Lastly, the output stage delivers the results and function of the input, leading in a creative Network Infrastructure Design in City Hall.

Network Architecture

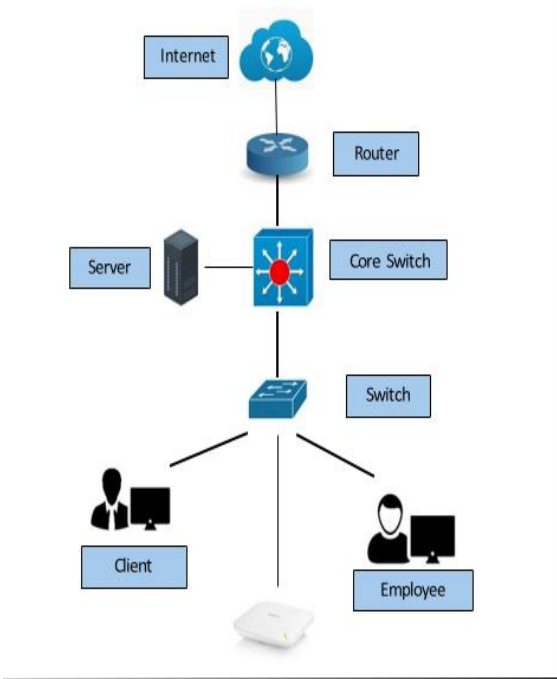


Figure 2. Network Architecture of Network Infrastructure Design in Cauayan City Hall

An architectural description is a description and representation of a system that is arranged in a way that allows for reasoning about the system's structures and actions.

Network Topology

A tree topology is an ideal choice for the network design of Cauayan City Hall due to its hierarchal structure and scalability. In a tree topology, network devices are organized into multiple levels, resembling to the structure of network layout. At the core of the network sits a central switch or router, representing the trunk, from which each of the department in Cauayan City Hall extend to connect to intermediate switches or routers on each floor or department, and finally, to end devices such as computer and printers.

VLAN ID

Table 2. VLAN ID

Devices	VLAN ID
Ground Floor	10
Second Floor	20
Third Floor	30
Access Point	40

City Human Resource	172.16.20.40/46	255.255.0.0
MNGT.OFFICE		
City General Services Office	172.16.20.47/53	255.255.0.0
City Accountant Office	172.16.20.54/60	255.255.0.0
City Accountant Staff	172.16.20.61/83	255.255.0.0
Mayor's Office	172.16.30.3/21	255.255.0.0
City Information and	172.16.30.22/40	255.255.0.0
Communication Office		

Table 3 . IP Address Scheme

DEPARTMENTS	IP ADDRESS	SUBNET MASK
City Treasurer Office	172.16.10.3/28	255.255.0.0
City Budget Office	172.16.10.29/34	255.255.0.0
City Budget Office Staffs	172.16.10.35/41	255.255.0.0
One Stop Office	172.16.10.42/47	255.255.0.0
BAC Office	172.16.10.48/54	255.255.0.0
BPLO	172.16.10.55/80	255.255.0.0
City Tourism Office	172.16.10.81/93	255.255.0.0
City Planning & Development Office	172.16.20.3/17	255.255.0.0
City Engineers Office	172.16.20.18/21	255.255.0.0
City Architect Office	172.16.20.22/30	255.255.0.0
Commission on Audit Office	172.16.20.31/39	255.255.0.0

3. Network Design Methodology

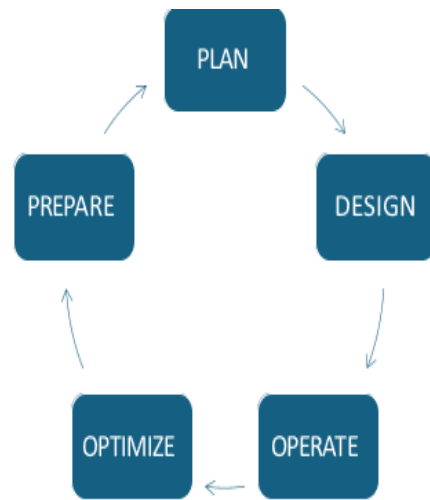


Figure 3. Network Design of Network Infrastructure Design in Cauayan City Hall

PHASE 1: PREPARE

In this phase, the researchers gathered requirements including knowing the exact location of where the network devices will be placed. Also, they determined any loopholes

and insufficiencies in their current network setup.

PHASE 2: PLAN

This phase involves the process of evaluating the sites where the network is intended to be deployed, as well as any pre-existing networks. This includes conducting a thorough gap analysis to determine whether the current system infrastructure, sites, and operational environment can accommodate the proposed system.

PHASE 3: DESIGN

This phase involves the researchers' gathering of requirements, which led the network design. The design is according to the determined initial requirements, utilizing any additional data gathered during network analysis and network audit, particularly when upgrading an existing network, as well as through discussions with department head and network users and finally, the utilization of cisco packet tracer to design the network and simulate in computing environment.

PHASE 4: OPERATE

This phase involves sustaining the health of the network through day-to-day operations, which includes ensuring high availability and minimizing expenses, and testing the proposed network design by using Cisco Packet Tracer for simulation.

PHASE 5: OPTIMIZE

This phase focuses specifically on enhancing system performance, provides a structured framework for managing the entire lifecycle of the network design, including optimization.

Respondents of the Study

To determine the required number of respondents, the researchers used a reliable method.

Table 8. Frequency Distribution of Respondents

Respondent	Frequency	Percentage
IT Experts	5	33%
ICT Coordinator	10	67%
Total	15	100%

Sampling Technique

This Study used stratified random sampling techniques to ensure an accurate and unbiased presentation of the population being studied.

$$n_i = f_i \times N_i$$

Where:

n_i = Sample size for stratum i.

f_i = Sampling fraction for stratum i ($f_i = n_i / N_i$).

N_i = Population size of stratum i.

4. FINDINGS OF THE ASSESSMENT

the current hardware used at City Hall, but several issues hinder its efficiency. The limited number of printers may not suffice for 193 computers, especially in high-demand departments, leading to delays and overuse. The router, with only 512MB of RAM, struggles to handle peak traffic volumes, and the absence of a centralized server necessitates

the costly use of 10 separate routers. Switches with only 5–10 ports restrict scalability and complicate network expansion, while varying computer specifications and the use of slower HDDs in some systems create performance gaps and inefficiencies. To address these issues, it is recommended to add more printers or centralized high-capacity models, upgrade to high-performance routers, and introduce a central server to reduce router dependence. Replacing low-port switches with higher-port models and standardizing switch brands will simplify management and support expansion. Additionally, upgrading computers to SSDs and standardizing specifications across departments may improve workflow efficiency. Implementing managed switches and layer 3 devices will streamline traffic flow, and regular hardware audits will ensure long-term reliability and performance.

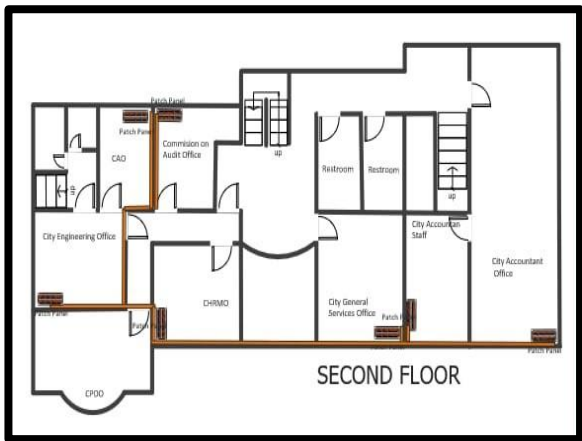
Metric	Original	Proposed
Printing Capacity	20	40
Router Traffic Handling	100	300
Switch Connectivity	10	20
Data Access Speed	100	10

In the original setup, printers were inadequate for high-demand departments, leading to delays. This proposal introduces centralized, high-capacity network printers, reducing queue times by 50%. Similarly, the current routers with 512MB RAM struggle with peak traffic, while the proposed upgrade to 2GB routers ensures smooth operation even during heavy usage, improving traffic handling by 3x. By replacing limited-port switches with higher-capacity models and standardizing hardware, this proposal not only resolves current connectivity bottlenecks but also simplifies future expansions and maintenance. These enhancements demonstrate clear, measurable improvements over the existing configuration.

Figure 8. Proposed Ground Floor Physical Layout

The figure shows the proposed physical layout of the Cauayan City Hall ground floor, ensuring that all departments are connected to each other for seamless communication and data monitoring.

Proposed Second Floor



The figure shows the proposed physical layout of the Cauayan City Hall second floor. Ensuring department are connected to each other for seamless communication and data monitoring.

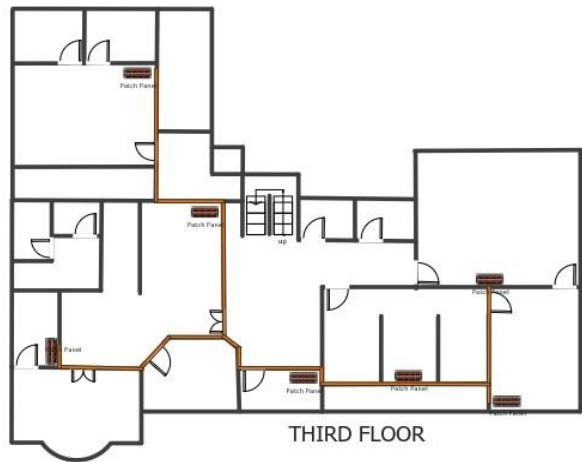


Figure 10. Proposed Third Floor Physical Layout

The figure illustrates the proposed physical layout of the third floor of Cauayan City Hall, designed to ensure that all departments are interconnected. This layout promotes seamless communication and efficient data monitoring by unifying the network infrastructure across all departments.

A tree topology is a more suitable alternative due to its hierarchical structure and scalability. In this design, a central router or switch serves as the "trunk" of the network, connecting to intermediate switches or routers that branch out to different departments or floors. This layered approach distributes the network load more effectively and reduces the risk of a single point of failure. Furthermore, it simplifies network expansion and management, making it easier to isolate and troubleshoot issues without disrupting the entire system.

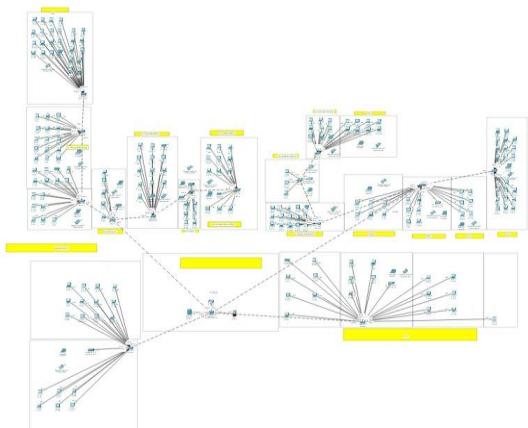


Figure 12. Existing Network Infrastructure Design using Cisco Packet Tracer

Figure 12 shows the proposed design using cisco packet tracer where researcher examine the use of tree topology instead on the original design star topology on their network map.

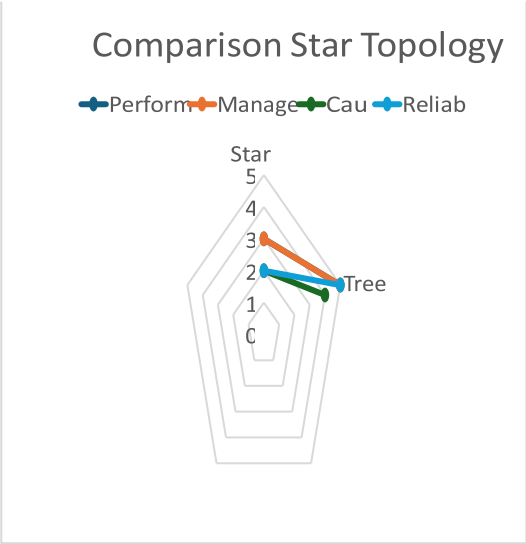


Figure 13. Existing Network Infrastructure Design using Cisco Packet Tracer

After the gathering of data and after comparing the findings, it clearly indicates that the tree topology offers significant advantages over the current star topology. By consolidating the network into a hierarchical structure, the tree topology reduces costs, enhances scalability, and improves overall performance. The risk of network-wide disruption is minimized, and management becomes more efficient. While the initial transition may require an investment in new equipment, the long-term savings and operational benefits outweigh the costs.

Network Speed/Performance

The current network infrastructure at City Hall, which allocates 100 Mbps per department, often experiences congestion, particularly in departments with a large number of connected devices. Additionally, excessive internet usage by some employees negatively impacts overall performance, causing slower

speeds for others. To address these challenges, it is recommended to upgrade departmental bandwidth to at least 1 Gbps, supported by a fiberoptic backbone for higher data transfer rates. Implementing Quality of Service (QoS) policies can prioritize critical traffic, while network segmentation through VLANs will help isolate departmental traffic and reduce congestion. Replacing outdated hardware with Gigabit or Multi-Gigabit network devices and deploying Wi-Fi 6 access points in high-density areas will further improve performance. Additionally, setting a bandwidth limit of 10 Mbps per PC using bandwidth management techniques, such as throttling and firewall rules will ensure fair usage and prevent misuse. These measures will optimize network speed, minimize congestion, and provide reliable connectivity for all departments.

Figure 14

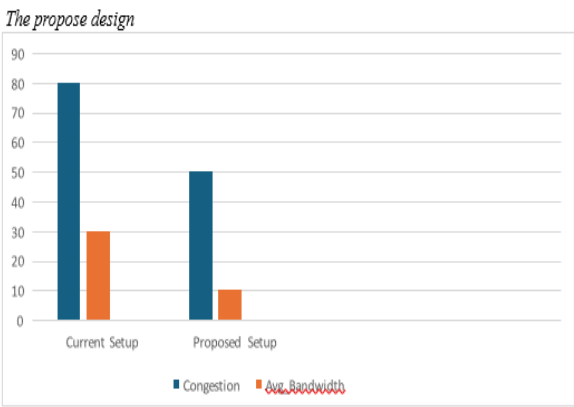


Figure 14. Result of Comparison between the Current Set-up and the Proposed

Based on the simulation, the figures clearly illustrate that the proposed network setup offers superior performance in managing congestion and maintaining average bandwidth. The enhanced configuration significantly reduces bottlenecks, ensuring a more even distribution of bandwidth across departments, especially in high-traffic areas. This improvement highlights the effectiveness of the

proposed upgrades, making the network more reliable and capable of supporting the demands of modern applications and increased device usage.

Network Security and Processes

Cauayan City Hall has also been experiencing data breaches, leading to the leak of sensitive private files containing the data they collect every day. This situation raises serious concerns about the security of the information entrusted to them by the citizens. The breach has not only compromised the privacy of individuals but has also damaged the trust between the public and the city hall, potentially exposing them to legal consequences and reputational harm. The current security measures rely heavily on Access Control Lists (ACLs) to regulate network traffic. While ACLs can provide a basic level of security by allowing or denying traffic based on IP addresses and port numbers, they have significant limitations that make them insufficient for addressing modern security threats and only evaluate the headers of packets, meaning they cannot analyze the contents of data being transmitted. This makes them ineffective against sophisticated attacks, such as malware embedded in data packets

Additionally, the absence of a firewall further compounds the problem. Without a firewall, there is no centralized mechanism to monitor and filter network traffic, leaving the entire network exposed to potential breaches. Firewalls also provide critical features like intrusion prevention, traffic logging, and advanced threat detection none of which are available with ACLs alone. Above all, Cauayan City Hall also uses unlicensed Microsoft office that may be one of the causes of breaches, unlicensed software does not receive official updates, leaving it vulnerable to exploits and cyberattacks.

Services

The researchers recommended optimizing the network infrastructure to improve the delivery of network services, ensuring both internal and operations and public services are supported efficiently. Given that the current monitoring tool is the Tenda G3, the researchers recommend enhancing its capabilities to optimize network performance and service delivery. The Tenda G3 is a useful tool, but integrating additional network monitoring software could provide more detailed insights, such as traffic analysis, real-time alerts, and performance metrics for better decision-making.

IP Addressing Structure

Table 13. Existing IP Address Scheme

DEPARTMENTS	IP ADDRESS	SUBNET MASK
City Treasurer Office	192.168.10.3/28	255.255.255.0
City Budget Office	192.168.10.29/34	255.255.255.0
City Budget Office Staffs	192.168.10.35/41	255.255.255.0
One Stop Office	192.168.20.42/47	255.255.255.0
BAC Office	192.168.30.42/47	255.255.255.0
BPLO	192.168.30.42/47	255.255.255.0
City Tourism Office	192.168.30.42/47	255.255.255.0
City Planning & Development Office	192.168.10.42/47	255.255.255.0

City Engineers Office	192.168.10.42/47	255.255.255.0
City Architect Office	192.168.10.42/47	255.255.255.0
Commission on Audit Office	192.168.20.42/47	255.255.255.0
City Human Resource MNGT.OFFICE	192.168.20.42/47	255.255.255.0
City General Services Office	192.168.20.42/47	255.255.255.0
City Accountant Office	192.168.30.42/47	255.255.255.0
City Accountant Staff	192.168.30.42/47	255.255.255.0
Mayor's Office	192.168.10.42/47	255.255.255.0
City Information and Communication Office	192.168.20.42/47	255.255.255.0

5. Network Infrastructure Assessment: Findings and Insights from the Questionnaire

The data collected through the evaluation of the existing network infrastructure is summarized below, as derived from the Network Infrastructure Design Questionnaire.

Table 15. Physical Layout

Table 15. Physical Layout

QUESTIONS	AVERAGE	DESCRIPTION
1. The designed network floor plan effectively supports the Cauayan City Hall building connectivity needs.	4.33	STRONGLY AGREE
2. The designed network floor plan adequately addresses the Cauayan City Hall building scalability requirements.	4.33	STRONGLY AGREE
3. The design of the network floor plan makes networking connections and equipment easier to access.	4.4	STRONGLY AGREE
4. The network floor plan design allows for easy scalability to accommodate future growth and changes.	4.33	STRONGLY AGREE
WEIGHTED AVERAGE	4.35	STRONGLY AGREE

The existing IP addressing scheme is insufficient for interdepartmental communication, as it prevents seamless connectivity between the departments. The current configuration fails to integrate the network across all areas, creating barriers to efficient data sharing and collaboration. A revised IP addressing plan is needed to ensure proper communication between all departments.

In Table 15, respondents strongly agree with the physical layout of the network, reflected in a weighted mean of 4.33. IT experts and ICT coordinators unanimously confirmed that the network floor plan meets the connectivity needs of the Cauayan City Hall building (mean: 4.33). The design also received high ratings for effectively addressing scalability requirements (mean: 4.4) and ensuring easy accessibility for networking connections and equipment (mean: 4.33). Furthermore, the design facilitates seamless scalability to accommodate future growth and modifications (mean: 4.35).

Table 16. Network Map

QUESTIONS	AVERAGE	DESCRIPTION
1. The network map is easy to understand and interpret.	4.4	STRONGLY AGREE
2. The network map is regularly updated to reflect changes in the network.	4.33	STRONGLY AGREE
3. The network map includes all network devices and their connections.	4.33	STRONGLY AGREE
WEIGHTED AVERAGE	4.35	STRONGLY AGREE

Referring to Table 14, respondents strongly agree on the effectiveness of the network map, as evidenced by a weighted mean of 4.4. Specifically, they find the map easy to comprehend and interpret (mean: 4.33) and appreciate its regular updates, which accurately reflect changes in the network (mean: 4.33). Furthermore, participants strongly agree that the network map is comprehensive, encompassing all network devices and their connections (mean:4.35).

Table 17. IP Address Structure

QUESTIONS	AVERAGE	DESCRIPTION
1. The IP addressing structure is designed to easily meet scalability and growth requirements in the future.	4.8	STRONGLY AGREE
2. Efficient IP address subnetting is applied to improve the overall performance of the network.	4.0	STRONGLY AGREE
3. The IP addressing structure supports efficient routing of network traffic.	4.6	STRONGLY AGREE
WEIGHTED AVERAGE	4.47	STRONGLY AGREE

Table 18. Network Security and Processes

QUESTIONS	AVERAGE	DESCRIPTION
1. The network security measures in place are adequate and effective.	4.2	STRONGLY AGREE
2. The network security measures protect against both internal and external.	4.4	STRONGLY AGREE
3. Network security do not unnecessarily hinder network performance or usability.	4.5	STRONGLY AGREE
WEIGHTED AVERAGE	4.37	STRONGLY AGREE

According to Table 18, respondents strongly agree (weighted mean: 4.2) that the network's security measures are both effective and sufficient. These measures are rated highly for protecting against internal and external threats (4.4), while maintaining network performance and usability (4.5). Baum and Dawes (2017) highlight that effective security systems integrate components that automatically extract and transfer security data between controllers, ensuring seamless operation and enhanced protection.

Table 19. Devices connected in the network

Criteria	Mean	Interpretation
Devices connected to the network communicate efficiently with each other.	4.2	STRONGLY AGREE
The network ensures secure connections for devices, protecting against unauthorized access or threats.	4.4	STRONGLY AGREE
Managing of network devices are effective and efficient.	4.5	STRONGLY AGREE
Weighted Mean	4.37	STRONGLY AGREE

In Table 19 shows that IT experts and ICT coordinators, as respondents, strongly agree (weighted mean: 4.83) with the presented miniature. They confirm its correctness in representing the actual Cauayan City Hall infrastructure (4.90) and the ideal alignment between cable connections in the miniature and Cisco Packet Tracer (4.80). Additionally, there is a shared acknowledgment of the miniature's effectiveness in showcasing well-managed cables, ensuring both neatness and safety (4.80).

Table 21. Result of Simulation

Questions	Mean	Interpretation
The routing path from the server to the client is accurately illustrates in the simulation.	4.70	STRONGLY AGREE
The routing path from the client to the server is accurately illustrates in the simulation.	4.90	STRONGLY AGREE
Data is securely transmitted between the client and server.	4.80	STRONGLY AGREE
The server handles multiple client requests efficiently.	4.80	STRONGLY AGREE
Weighted Mean	4.80	STRONGLY AGREE

Tables 20 shows that respondents Strongly Agree with a weighted mean of 4.80. Particularly, respondents strongly agree that the simulation accurately reflects both the server-to-client (4.70) and client-to-server (4.90) routing pathways. Furthermore, there is a strong confidence in the secure transfer of data (4.80) and the server's effective

Cost-Benefit Analysis

This section examines the potential cost savings and various benefits associated with implementing a star topology design.

Material Cost:

This section evaluates the financial implications of implementing the network infrastructure at Cauayan City Hall, focusing on the cost-effectiveness of the proposed network design.

Table 23. Hardware Cost of Network Infrastructure Design in Cauayan City Hall

Material	Unit	Price	Total Price
Router	1	₱ 8,349.04	₱ 8,349.04
Managed Switch TP-Link TL-SG108PE	1	₱15,846.06	₱15,846.06
Switch Cisco Catalyst 2950 24-PORT 10/100 Fe	14	₱8,102.00	₱43,428.00
Access point/ WIFI Ubiquiti Networks UniFi AP LR WiFi PoE Wireless Access Point	14	₱ 6,999.00	₱97,986.00
UTP Cable	2,000m	₱ 65.00	₱120,000.00
RJ45 Cat6 100 pcs per box	3	₱ 964.98	₱2,894.94
Patch Outlet/ RJ45 double socket	13	₱ 290.02	₱3,770.26
Server TERRAMASTER T9-450 9Bay NAS Storage - High Speed Network Attached Storage with Atom C3558R Quad-core CPU, 8GB DDR4 Memory, Dual SFP+ 10GbE Interfaces, Dual 2.5GbE Ports, NAS Server (Diskless)	1	₱ 26,799.00	₱ 260,799.00
Total			₱623,072.36

Benefits:

The network infrastructure of Cauayan City Hall, built with a tree topology provides several significant benefits to Cauayan City Hall. These benefits include:

- The hierarchical structure allows easy expansion by adding new branches without disrupting the existing network.
- Combines the advantages of both star and bus topologies, minimizing redundant connections and hardware expenses.
- Centralizes control enables easier troubleshooting and network management.
- Aids in identifying additional hardware to be needed in performing network design.

Summary

Currently, the Cauayan City Hall has 10 routers in each department and offices, and it has a star topology network setup. However, this configuration presents several challenges, including limited scalability, inefficient performance, and restricted access to critical online resources. Additionally, the absence of centralized network control hinders effective traffic management, optimization, and troubleshooting. Study aims to address these problems by providing a strategic design and solution.

The functionality and practicality of the proposed network infrastructure design for the

Cauayan City Hall were evaluated using the weighted mean and descriptive interpretation methods. The assessment involved 15 IT specialists who participated in the evaluation process.

The suggested network infrastructure design uses Cisco Packet Tracer simulation to prioritize planning and designing for efficient networking development, based on the PPDIOO paradigm. This aligned approach covers five of the six phases (Prepare, Plan, Design, Operate, and Optimize) with the exception of the implementation step. This makes it possible to conduct a focused and substantial simulation-based study on network modeling.

The proposed network infrastructure design leverages Cisco Packet Tracer simulation to prioritize effective planning and design for efficient network development, following the PPDIOO paradigm. This approach encompasses five of the six phases—Prepare, Plan, Design, Operate, and Optimize—excluding only

the implementation phase. This methodology enables a focused and comprehensive simulation-based study of network modeling.

The study examines the existing network infrastructure at Cauayan City Hall and proposes an optimized design aimed at addressing several issues impacting network performance. The current hardware setup includes 12 printers, 10 routers, 15 switches, and 193 computers, each with varying specifications. While the system supports the daily operations of the City Hall, several hardware limitations, including insufficient printers, under-performing routers, and outdated devices, are identified. Additionally, the existing star topology proves inefficient, leading to congestion and higher operational costs. To enhance performance, the study

proposes a shift to a tree topology, with upgrades to hardware such as high-capacity routers, switches, and centralized printers, alongside a more robust network layout. Furthermore, the use of VLANs, port security, and Wi-Fi 6 is recommended for improving both network security and connectivity.

The study utilized a custom-developed Network Infrastructure Design Questionnaire (NIDQ) to assess the feasibility and usability of the proposed network infrastructure. IT Experts and ICT Coordinators evaluated various aspects of the design, with the majority expressing strong agreement on key components: Physical Layout (4.33), Network Map (4.35), IP Address Structure (4.70), Network Security and Processes (4.49), Devices Connected in the Network (4.35), Miniature (4.64), and Simulation Results (4.72). The overall evaluation reflected a very positive reception, resulting in a weighted grand mean of 4.41.

Conclusion

The proposed hardware upgrades are designed to address the various limitations currently observed in the infrastructure at City Hall, aiming to significantly improve network efficiency, scalability, and overall performance. At present, the network struggles with several bottlenecks, including a limited number of printers, routers with insufficient capacity, and outdated switches. These limitations result in frequent delays, network congestion, and difficulties when scaling the infrastructure to accommodate the growing demands of City Hall's operations. The primary objective of the proposed upgrades is to modernize the entire network system, enhance performance, and support future expansion in a more efficient and cost-effective manner.

One of the key components of the proposal is the introduction of centralized, high capacity network printers. Currently, the

network printers are spread across multiple departments, causing delays due to heavy usage and inefficient management. By consolidating the printers into a single, high-capacity unit, the proposal aims to reduce printer queue times by up to 50%. This change will enable faster printing processes, reduce waiting times, and improve productivity across departments, particularly for tasks involving large print volumes. In addition to the printers, the proposal includes upgrading the routers to models with at least 2GB of RAM, which will significantly improve their ability to handle peak traffic loads and support the growing demand for internet access in a high-traffic environment. The upgraded routers will enable a much smoother network experience, improving connectivity and reducing latency by threefold during periods of heavy usage.

Further improvements will be made to the network's switches. The current switches are low-port models that are no longer suitable for the expanding number of connected devices in City Hall. The proposal calls for replacing these with higher-capacity, modern switches capable of managing larger volumes of data. These switches will facilitate better data transmission speeds, increase network stability, and provide additional scalability, allowing the network to grow without requiring significant further investment in hardware. With these upgraded switches in place, the network will be better equipped to handle future expansions without running into performance issues.

Additionally, the proposal suggests implementing a centralized server for managing data storage and resource sharing across departments. This centralized approach will not only streamline data access but also reduce the dependence on multiple routers. By consolidating data and applications onto a single server, network management becomes more straightforward, and troubleshooting issues becomes less complicated. Centralized storage will also enable more efficient data

backup processes and provide a more secure environment for sensitive City Hall data.

Finally, the proposal includes upgrading computers used across various departments Solid-State Drives (SSDs), which are known for their speed, durability, and energy efficiency. The upgrade will significantly improve responsiveness of applications, speed up retrieval times, and extend the lifespan of computers. Furthermore, the proposal calls for standardizing the computer specifications across departments to ensure consistency and ease of maintenance. By ensuring that all computers are equipped with similar specifications, troubleshooting becomes easier, and employees experience uniform performance levels, contributing to a smoother workflow.

Recommendation

Based on the results of the study, the researchers recommended the following:

1. Future researchers may investigate the impact of 5G technology on educational network infrastructures, exploring how faster and more reliable connected networks could influence the design and functionality of future systems.
2. Future Researchers are encouraged to explore emerging technologies in network infrastructure to ensure that proposed designs remain flexible and capable of incorporating future advancements in the field.

Future researchers are recommended to conduct in-depth studies on the scalability of network infrastructure designs to accommodate potential expansion needs, ensuring that the framework can support future growth and adapt to changes.

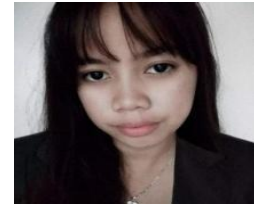
REFERENCES

- Afrin, T., & Yodo, N. (2019). Resilience-Based Recovery Assessments Of Networked Infrastructure Systems Under Localized Attacks. *Infrastructures*, 4(1), 11.
<https://doi.org/10.3390/Infrastructures4010011>
- Ahmed, A. H., & Al-Hamadani, M. N. A. (2021). Designing a Secure Campus Network And Simulating It Using Cisco Packet Tracer. *Indonesian Journal Of Electrical Engineering And Computer Science*, 23(1), 479.
<https://doi.org/10.11591/Ijecs.v23.i1.Pp479-489>
- Ahmad, A. H. (2022). Challenges And Effectiveness Of Using Cisco Packet Tracer Simulator In Learning Vpn Connection (Virtual Private Network). *International Journal Of Modern Education*, 4(14), 21–30.
<https://doi.org/10.35631/Ijmoec.414003>
- Alaoui, I. E., & Gahi, Y. (2020). Network Security Strategies In Big Data Context. *Procedia Computer Science*, 175, 730–736.
<https://doi.org/10.1016/j.procs.2020.07.108>

Annigeri1, S ., Chauhan, A ., & Chhatlan, J. (2020). Use Of Virtual Lans For Network Segmentation And Organization. *International Journal Of Research In Engineering, Science And Management (Ijresm)*.

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Azamuddin, W. M. H., Hassan, R., Aman, A. H. M., Hasan, M. K., & Al-Khaleefa, A. S. (2020). Quality Of Service (Qos) Management For Local Area Network (Lan) Using Traffic Policy Technique To Secure Congestion. *Computers*, 9(2), 39.
<https://doi.org/10.3390/Computers9020039>

Basu, D. (2019). Network Infrastructure Vulnerabilities And Its Mitigation: A Review.

Cerdà-Alabern, L., Iuhász, G., & Gemmi, G. (2023). Anomaly Detection For Fault Detection In Wireless Community Networks Using Machine Learning. *Computer Communications*, 202, 191–203.
<https://doi.org/10.1016/j.comcom.2023.02.019>

Cherifi, H., Palla, G., Szymanski, B. K., & Lu, X. (2019). On Community Structure In Complex Networks: Challenges And Opportunities. *Applied Network Science*, 4(1).
<https://doi.org/10.1007/s41109-019-0238-9>

OBJECTIVE:

To acquire valuable knowledge and skills to complement those that I have learned from school in an actual job environment. In return, I offer my service and determination to be an asset to your company throughout the duration of my training period.

Education

Tertiary: Bachelor of Science in Information Technology
Isabela State University
(2021 – Present)

Secondary:

Cauayan City National High School

Turayong, Cauayan City, Isabela

(2015-2020)

Elementary:

Cauayan West Central School

SKILLS

- Oriented in Microsoft Office such as:
 - MS Word
 - PowerPoint
 - Excel
- KNOWLEGABLE PHP and HTML
- MYSQL
- NETWORK CONFIGURATION
- COMMUNICATION SKILLS

Primary

Divilacan National
High School
(2015-2019)
Divilacan Central
School
Dimapula, Divilacan
Isabela
(2009-2015)

MICHAEL JOHN S. MECATE

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PERSONAL DATA

Place of Birth	City of Ilagan Isabela
Date Of Birth	February 10, 2003
Age	21
Address	Dilakit, Divilacan
Sex	Male
Nationality	Filipino

Tertiary	Bachelor of Science in Information Technology Isabela State University Calamagui 2 nd , Ilagan City, Isabela (2025-Present)
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Secondary	Isabela National High School San Vicente, Ilagan City Isabela
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